Roll No.....

Mid Semester Examination 2025

Name of the Program: M.Tech (ECE) | *Semester: I*
Name of the Course: Biophotonics | *Course Code:* BEM 104

Time: 90 minutes | Maximum Marks: 50

Note:
(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

Q1. (10 marks) **CO-01**

(a) Discuss the cell classification with structure and classification.

OR

(b) Discuss the protein classification and their functioning.

Q2. (10 marks) **CO-01**

(a) Explain the types of tissues and their functioning.

OR

(b) Explain the principle of photo-excitation. How is it used in biomedical optical diagnostics?

Q3. (10 marks) **CO-02**

(a) What is spectroscopy? Discuss different types of molecular spectroscopy used in biomedical applications.

OR

(b) Describe the working principle of UV/VIS absorption spectroscopy. Discuss its biomedical applications.

Q4. (10 marks) **CO-02**

(a) Describe the principle and instrumentation of FTIR spectroscopy. Mention its advantages in biomedical research.

OR

(b) What is Raman spectroscopy? Compare spontaneous Raman with Resonance Raman and SERS techniques.

Q5. (10 marks) **CO-01 & CO-02**

(a) Explain the principle of fluorescence spectroscopy. How is it used in biological imaging and diagnostics?

OR

(b) Describe the different types of Raman spectroscopy and their relevance in biomedical optics.